(5) The *component functions* f^i: $\mathbb{R}^n \to \mathbb{R}$ of a function f: $\mathbb{R}^n \to \mathbb{R}^m$ are defined by

$$f(\mathbf{x}) = (f^1(\mathbf{x}), f^2(\mathbf{x}), \ldots, f^m(\mathbf{x})).$$

Explain why

$$D_{\mathbf{u}}f(\boldsymbol{\xi}) = (D_{\mathbf{u}}f^1(\boldsymbol{\xi}), D_{\mathbf{u}}f^2(\boldsymbol{\xi}), \ldots, D_{\mathbf{u}}f^m(\boldsymbol{\xi})).$$

(6) Prove that the function f: $\mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x_1, x_2) = \begin{cases} \dfrac{x_1^2 x_2}{x_1^4 + x_2^2} & (x_1, x_2) \neq (0, 0) \\ 0 & (x_1, x_2) = (0, 0) \end{cases}$$

has directional derivatives in every direction at (0, 0) but is *not* continuous at (0, 0). [Hint: example 18.43.] Does f have a local maximum or minimum at (0, 0)?

19.12 Differentiable functions

Consider a function f: $A \to \mathbb{R}^m$ where A is a set which contains each $\mathbf{x} \in \mathbb{R}^n$ satisfying $\|\mathbf{x} - \boldsymbol{\xi}\| < \Delta$. What should it mean to say that f is differentiable at $\boldsymbol{\xi}$? We dealt with the case $m = n = 1$ in §10.1. Our definition required the existence of the limit

$$l = \lim_{x \to \xi} \frac{f(x) - f(\xi)}{x - \xi} = \lim_{h \to 0} \frac{f(\xi + h) - f(\xi)}{h}.$$

This may be rewritten as

$$f(x) - f(\xi) - l(x - \xi) = o(x - \xi) \qquad (x \to \xi)$$

or as

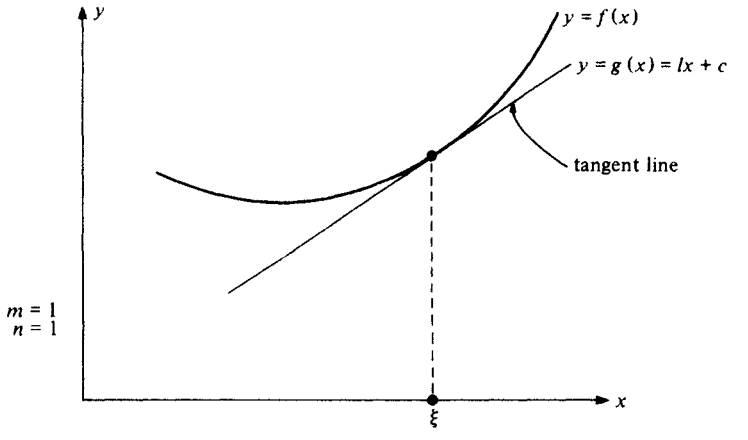
$$f(\xi + h) - f(\xi) - lh = o(h) \qquad (h \to 0)$$

where $o(h)$ denotes a quantity which tends to zero when divided by h. The number l, of course, is called the derivative of f at ξ and denoted by $f'(\xi)$.

An affine function g: $\mathbb{R} \to \mathbb{R}$ is defined by an equation of the form $g(x) = lx + c$. (See example 18.24(i).) We can therefore restate the definition yet again as being the requirement that there exist an affine function g: $\mathbb{R} \to \mathbb{R}$ with $g(\xi) = f(\xi)$ such that

$$f(x) - g(x) = o(x - \xi) \qquad (x \to \xi)$$

The geometric interpretation of this condition is that the graphs $y = f(x)$ and $y = g(x)$ are tangent where $x = \xi$.



The graph of an affine function is a straight line and so our analytic definition of differentiability corresponds to the geometric idea that it be possible to draw a tangent line to the graph of f where $x = \xi$.

We are now in a position to define differentiability in the general case when m and n are not necessarily equal to one. We say that $f\colon A \to \mathbb{R}^m$ is *differentiable* at $\boldsymbol{\xi}$ if and only if there exists an affine function $g\colon A \to \mathbb{R}^m$ with $g(\boldsymbol{\xi}) = f(\boldsymbol{\xi})$ such that

$$f(\mathbf{x}) - g(\mathbf{x}) = o(\mathbf{x} - \boldsymbol{\xi}) \qquad (\mathbf{x} \to \boldsymbol{\xi}).$$

One cannot, of course, divide by a vector and $o(\mathbf{x} - \boldsymbol{\xi})$ in this equation denotes a quantity which tends to zero when divided by the real number $\|\mathbf{x} - \boldsymbol{\xi}\|$. The criterion can therefore be rewritten as

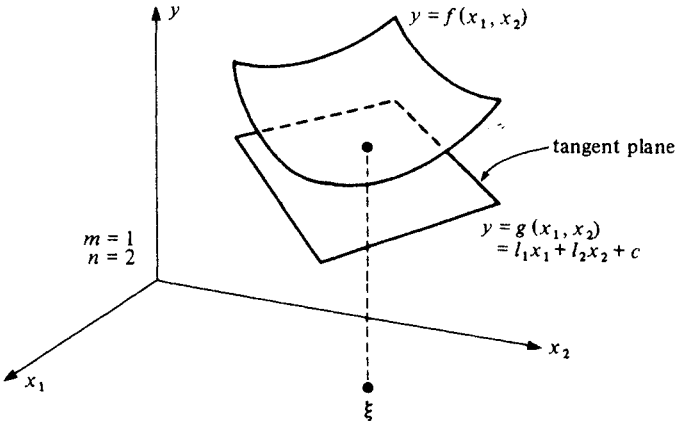
$$\lim_{\mathbf{x} \to \boldsymbol{\xi}} \frac{\|f(\mathbf{x}) - g(\mathbf{x})\|}{\|\mathbf{x} - \boldsymbol{\xi}\|} = 0.$$

As in the $m = n = 1$ case, the geometric interpretation is that the graphs $\mathbf{y} = f(\mathbf{x})$ and $\mathbf{y} = g(\mathbf{x})$ are tangent at $\mathbf{x} = \boldsymbol{\xi}$. The graph of an affine function is a *flat*. (See §18.18 and exercise 18.25(4).) The definition therefore corresponds to the geometric idea that it be possible to draw a *tangent flat* to the graph of f where $\mathbf{x} = \boldsymbol{\xi}$.

In the case $m = n = 1$, a tangent flat at $x = \xi$ is simply a straight line as illustrated above. In example 18.24(ii) we saw that an affine function $g\colon \mathbb{R}^2 \to \mathbb{R}^1$ is defined by an equation of the form

$$y = l_1 x_1 + l_2 x_2 + c$$

and its graph is therefore a plane in $\mathbb{R}^3$. In the case $m = 1$ and $n = 2$ it therefore follows that a tangent flat at $(x_1, x_2) = (\xi_1, \xi_2)$ is a plane as in the diagram below.

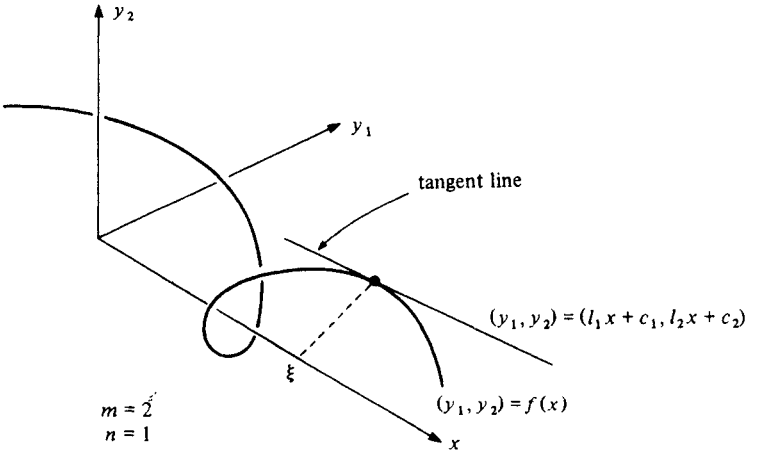


In example 18.24(iii) we saw that an affine function g: $\mathbb{R}^1 \to \mathbb{R}^2$ is defined by two equations of the form

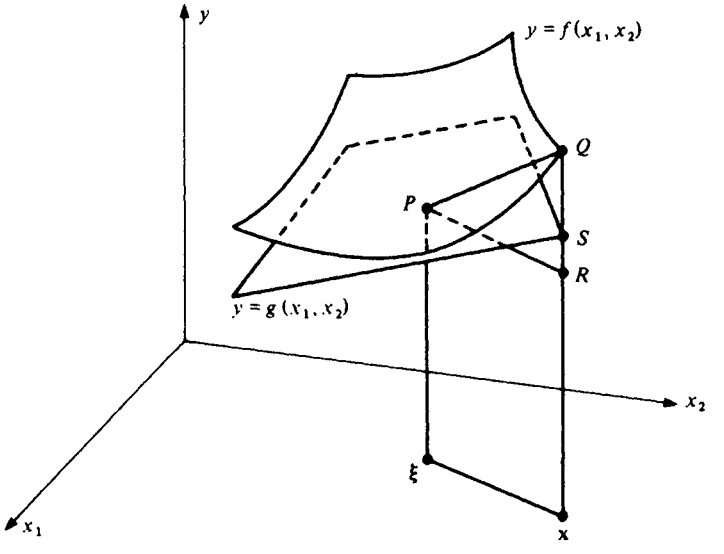
$$y_1 = l_1x_1 + c_1$$

$$y_2 = l_2x_2 + c_2$$

and its graph is therefore a line in $\mathbb{R}^3$. In the case $m = 2$ and $n = 1$, it therefore follows that a tangent flat at $x = \xi$ is a line as indicated below.



In §10.1 we gave a geometric justification for our identification of the idea of differentiability with the existence of a tangent line in the case $m = n = 1$. We now give a similar justification for the case $m = 1$ and $n = 2$.



In the diagram, Q lies on the surface $y = f(x_1, x_2)$ and S on the surface $y = g(x_1, x_2) = l_1 x_1 + l_2 x_2 + c$. The point R is at the same height as P. The condition for $y = g(x_1, x_2)$ to be tangent to $y = f(x_1, x_2)$ at P is that

$$\lim_{\mathbf{x} \to \xi} \left\{ \frac{QR}{PR} - \frac{SR}{PR} \right\} = 0.$$

But $PR = \|\mathbf{x} - \boldsymbol{\xi}\|$ and $|QR - SR| = |QS| = |f(\mathbf{x}) - g(\mathbf{x})|$. Our geometrically derived criterion therefore reduces to the condition

$$\lim_{\mathbf{x} \to \xi} \frac{|f(\mathbf{x}) - g(\mathbf{x})|}{\|\mathbf{x} - \boldsymbol{\xi}\|} = 0$$

which is the definition of differentiability in the case $m = 1$ and $n = 2$.

19.13 Theorem Let $f: A \to \mathbb{R}^m$ where A contains all $\mathbf{x} \in \mathbb{R}^n$ which satisfy $\|\mathbf{x} - \boldsymbol{\xi}\| < \Delta$. If f is differentiable at $\boldsymbol{\xi}$, then f is continuous at $\boldsymbol{\xi}$.

Proof From the definition of differentiability there exists an affine function $g: \mathbb{R}^n \to \mathbb{R}^m$ with $g(\boldsymbol{\xi}) = f(\boldsymbol{\xi})$ such that

$$\lim_{\mathbf{x} \to \xi} \frac{\|f(\mathbf{x}) - g(\mathbf{x})\|}{\|\mathbf{x} - \boldsymbol{\xi}\|} = 0$$

It follows (as in theorem 10.6) that

$$\|f(\mathbf{x}) - g(\mathbf{x})\| \to 0 \text{ as } \mathbf{x} \to \boldsymbol{\xi}.$$

But g is necessarily continuous at $\boldsymbol{\xi}$ (example 18.34). Hence

$$\begin{aligned}\|f(\mathbf{x})-f(\boldsymbol{\xi})\| &= \|f(\mathbf{x})-g(\boldsymbol{\xi})\| \\ &= \|f(\mathbf{x})-g(\mathbf{x})+g(\mathbf{x})-g(\boldsymbol{\xi})\| \\ &\leqslant \|f(\mathbf{x})-g(\mathbf{x})\|+\|g(\mathbf{x})-g(\boldsymbol{\xi})\| \\ &\to 0+0 \text{ as } \mathbf{x}\to\boldsymbol{\xi}.\end{aligned}$$

Thus $f(\mathbf{x})\to f(\boldsymbol{\xi})$ as $\mathbf{x}\to\boldsymbol{\xi}$ and so f is continuous at $\boldsymbol{\xi}$.

19.14 Derivatives

In this section we shall need the ideas introduced in §18.22. In particular, we shall find it convenient to replace $\mathbf{x}$ and $\mathbf{y}$ by *column* vectors. To emphasise this substitution we shall write x and y where previously we have written $\mathbf{x}$ and $\mathbf{y}$.

An affine function $g\colon \mathbb{R}^n\to\mathbb{R}^m$ is defined by an equation of the form

$$y = g(x) = Lx + c$$

where L is an $m\times n$ matrix and c is an $m\times 1$ column vector. If $g(\xi)=f(\xi)$, we may rewrite this equation in the form

$$y = g(x) = L(x-\xi)+f(\xi).$$

It follows that f is differentiable at ξ if and only if there exists an $m\times n$ matrix L such that

$$f(x)-f(\xi)-L(x-\xi) = o(x-\xi) \qquad (x\to\xi)$$

or, what is the same thing,

$$f(\xi+h)-f(\xi)-Lh = o(h) \qquad (h\to 0).$$

These equations are identical in form to the second and third equations of §19.12 except that L replaces l. This correspondence is very significant but it must not be forgotten that, in the current equations, the small letters all represent column *vectors* and L represents a *matrix* whereas in §19.12 all the letters in the equations represent real numbers. Note in particular that the real number l in §19.12 is the derivative $f'(\xi)$ of f at ξ. This leads us to a natural definition for the derivative of a differentiable function in the general case. If f is differentiable at ξ, we define its *derivative* at ξ by

$$f'(\xi) = L.$$

Sometimes the notation $Df(\xi)$ is used instead.

The derivative of f at ξ is therefore an $m\times n$ *matrix* in the general case. A number of advantages follow from the use of this definition. The tangent flat at ξ, for example, is given by the equation

$$y = f(\xi)+f'(\xi)(x-\xi)$$

exactly as in the case $m=n=1$.

How is the derivative of a vector function calculated? We begin to answer this question with the following theorem which connects the idea of a derivative introduced above with that of a directional derivative introduced earlier.

19.15 Theorem Suppose that $f: A \to \mathbb{R}^m$ where A contains all $x \in \mathbb{R}^n$ satisfying $\|x - \xi\| < \Delta$. If f is differentiable at ξ, then all directional derivatives of f at ξ exist and

$$D_u f(\xi) = f'(\xi)u.$$

Proof We have that

$$\frac{\|f(x) - f(\xi) - f'(\xi)(x - \xi)\|}{\|x - \xi\|} \to 0$$

as x approaches ξ and, in particular, as x approaches ξ along the straight line through ξ in the direction u. Writing $x = \xi + tu$, we obtain that

$$\lim_{t \to 0} \left\| \frac{f(\xi + tu) - f(\xi)}{t} - f'(\xi)u \right\| = 0$$

and hence $D_u f(\xi) = f'(\xi)u$.

With the assumptions of the preceding theorem, $f(x)$ denotes an $m \times 1$ column vector. The formula

$$f(x) = \begin{pmatrix} f^1(x) \\ f^2(x) \\ \vdots \\ f^m(x) \end{pmatrix}$$

therefore determines m real-valued functions $f^1, f^2, \ldots, f^m$ which we call the *component* functions of f. In the next theorem we explain how the entries in the matrix

$$f'(\xi) = \begin{pmatrix} l_{11} & l_{12} \ldots l_{1n} \\ l_{21} & l_{22} \ldots l_{2n} \\ \vdots & \\ l_{m1} & l_{m2} \ldots l_{mn} \end{pmatrix}$$

can be calculated from the partial derivatives of these component functions. (An incidental corollary is that the derivative of a differentiable function is uniquely defined.)

19.16 Theorem With the assumptions of the preceding theorem,

$$f'(\xi) = \begin{pmatrix} f_1^1(\xi) & f_2^1(\xi) \dots f_n^1(\xi) \\ f_1^2(\xi) & f_2^2(\xi) \dots f_n^2(\xi) \\ \vdots & \\ f_1^m(\xi) & f_2^m(\xi) \dots f_n^m(\xi) \end{pmatrix}$$

Proof We shall only prove that $l_{12} = f_2^1(\xi)$. We take $u = (0, 1, 0, \dots, 0)$. Then, by exercise 19.11(5),

$$D_u f(\xi) = \begin{pmatrix} D_u f^1(\xi) \\ D_u f^2(\xi) \\ \vdots \\ D_u f^m(\xi) \end{pmatrix} = \begin{pmatrix} f_2^1(\xi) \\ f_2^2(\xi) \\ \vdots \\ f_2^m(\xi) \end{pmatrix}$$

Also

$$f'(\xi)u = \begin{pmatrix} l_{11} & l_{12} \dots l_{1n} \\ l_{21} & l_{22} \dots l_{2n} \\ \vdots & \\ l_{m1} & l_{m2} \dots l_{mn} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} l_{12} \\ l_{22} \\ \vdots \\ l_{m2} \end{pmatrix}$$

Comparing the first co-ordinates of each column and appealing to theorem 19.15, we obtain that $l_{12} = f_2^1(\xi)$.

If $y = f(x)$, it is natural to use the notation

$$\frac{\mathrm{d}y}{\mathrm{d}x}$$

for the derivative of a vector function (with our usual reservations about the ambiguous use of x). The formula of theorem 19.16 then takes the pleasing and easily remembered form

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \begin{pmatrix} \dfrac{\partial y_1}{\partial x_1} & \dfrac{\partial y_1}{\partial x_2} \dots \dfrac{\partial y_1}{\partial x_n} \\ \dfrac{\partial y_2}{\partial x_1} & \dfrac{\partial y_2}{\partial x_2} \dots \dfrac{\partial y_2}{\partial x_n} \\ \vdots & \\ \dfrac{\partial y_m}{\partial x_1} & \dfrac{\partial y_m}{\partial x_2} \dots \dfrac{\partial y_m}{\partial x_n} \end{pmatrix}$$

We have seen that the derivative of a differentiable vector function f is a matrix whose entries consist of all partial derivatives of the co-ordinates of f. It is important, however, to bear in mind that the mere existence of this matrix does *not* guarantee that f is differentiable. The function of exercise 19.11(6) provides a suitable example. All its partial derivatives exist at **O** but the function

is not even continuous at **O** and hence is certainly not differentiable at **O** (theorem 19.13). Thus, although the matrix of partial derivatives can be written down, it does not help very much since the function simply does not have a tangent flat at **O**.

How can one tell whether a vector function is differentiable? The following theorem provides a criterion which is adequate for most practical purposes.

19.17 Theorem Suppose that all partial derivatives of f are *continuous* at ξ. Then f is differentiable at ξ.

Proof To avoid cumbersome notation, we consider only the case $m = 1$ and $n = 2$. From the mean value theorem (11.6) we have that

$$\begin{aligned} f(x_1, x_2) - f(\xi_1, \xi_2) &= \{f(x_1, x_2) - f(\xi_1, x_2)\} + \{f(\xi_1, x_2) - f(\xi_1, \xi_2)\} \\ &= f_1(u_1, x_2)(x_1 - \xi_1) + f_2(\xi_1, u_2)(x_2 - \xi_2) \end{aligned}$$

where u_1 lies between x_1 and ξ_1 while u_2 lies between x_2 and ξ_2. Thus $|u_1 - \xi_1| < |x_1 - \xi_1| \leqslant \|x - \xi\|$ and so $u_1 \to \xi_1$ as $x \to \xi$. It follows that $(u_1, x_2) \to (\xi_1, \xi_2)$ as $x \to \xi$. Similarly $(\xi_1, u_2) \to (\xi_1, \xi_2)$ as $x \to \xi$. Since f_1 and f_2 are continuous at ξ, we deduce that

$$f_1(u_1, x_2) \to f_1(\xi_1, \xi_2) \text{ as } x \to \xi,$$

$$f_2(\xi_1, u_2) \to f_2(\xi_1, \xi_2) \text{ as } x \to \xi$$

(a conclusion which requires the vector analogue of theorem 8.17).

We now show that

$$f(x) - f(\xi) - L(x - \xi) = o(x - \xi) \quad (x \to \xi)$$

with $L = (f_1(\xi), f_2(\xi))$. This follows from the fact that

$$\begin{aligned} &|f(x) - f(\xi) - f_1(\xi)(x_1 - \xi_1) - f_2(\xi)(x_2 - \xi_2)| \\ &\quad = |(f_1(u_1, x_2) - f_1(\xi_1, \xi_2))(x_1 - \xi_1) + (f_2(\xi_1, u_2) \\ &\quad - f_2(\xi_1, \xi_2))(x_2 - \xi_2)| \leqslant E\|x - \xi\| \end{aligned}$$

by the Cauchy–Schwarz inequality, where

$$\begin{aligned} E &= \{(f_1(u_1, x_2) - f_1(\xi_1, \xi_2))^2 + (f_2(\xi_1, u_2) - f_2(\xi_1, \xi_2))^2\}^{1/2} \\ &\to \{0^2 + 0^2\}^{1/2} = 0 \text{ as } x \to \xi. \end{aligned}$$

19.18 Examples

(i) Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ be defined by $f(x_1, x_2) = x_1^2 x_2 + x_2^3 x_1$. Then

$$\left.\begin{aligned} f_1(x_1, x_2) &= 2x_1 x_2 + x_2^3 \\ f_2(x_1, x_2) &= x_1^2 + 3x_2^2 x_1. \end{aligned}\right\}$$

These partial derivatives are polynomials and hence are continuous everywhere (proposition 18.38). Thus f is differentiable at every point. The derivative at the point (2, 1) is the 1×2 matrix

$$\begin{aligned} f'(2, 1) &= (f_1(2, 1), f_2(2, 1)) \\ &= (5, 10). \end{aligned}$$

Various facts about the behaviour of f at (2, 1) may be deduced from this information. For example, the slope (or rate of increase) of f at the point (2, 1) in the direction (4, 3) is given by

$$D_u f(1, 2)$$

where u is the unit vector which points in the same direction as (4, 3). From theorem 19.15 we have that

$$\begin{aligned} D_u f(1, 2) &= f'(1, 2)u \\ &= (5, 10)\begin{pmatrix} 4/5 \\ 3/5 \end{pmatrix} = 4 + 6 = 10. \end{aligned}$$

We can also obtain the equation of the tangent plane to the surface $y = x_1^2 x_2 + x_2^3 x_1$ where $(x_1, x_2) = (1, 2)$. This is given by writing $(\xi_1, \xi_2) = (1, 2)$ in the formula

$$y = f(\xi) + f'(\xi)(x - \xi).$$

This yields

$$y = 6 + (5, 10)\begin{pmatrix} x_1 - 1 \\ x_2 - 2 \end{pmatrix}$$

i.e.

$$y - 6 = 5(x_1 - 1) + 10(x_2 - 2).$$

(ii) Let $f\colon \mathbb{R}^3 \to \mathbb{R}$ be defined by $f(x_1, x_2, x_3) = x_1 x_2^2 + x_2 x_3^3 + x_3 x_1^4$. Then

$$\left.\begin{aligned} f_1(x_1, x_2, x_3) &= x_2^2 + 4x_3 x_1^3 \\ f_2(x_1, x_2, x_3) &= 2x_1 x_2 + x_3^3 \\ f_3(x_1, x_2, x_3) &= 3x_2 x_3^2 + x_1^4. \end{aligned}\right\}$$

These partial derivatives are continuous at every point and hence f is differentiable everywhere. The derivative at (1, 1, 1) is the 1×3 matrix

$$f'(1, 1, 1) = (5, 3, 4).$$

The slope (or rate of increase) of f at (1, 1, 1) in the direction (2, 1, 2) is equal to

$$f'(1, 1, 1)u = (5, 3, 4)\begin{pmatrix} 2/3 \\ 1/3 \\ 2/3 \end{pmatrix} = 7$$

(where u is the unit vector which points in the same direction as $(2, 1, 2)$).

The tangent flat to $y = x_1x_2^2 + x_2x_3^3 + x_3x_1^4$ where $(x_1, x_2, x_3) = (1, 1, 1)$ is the hyperplane

$$y = 3 + (5, 3, 4)\begin{pmatrix} x_1 - 1 \\ x_2 - 1 \\ x_3 - 1 \end{pmatrix}$$

i.e.

$$y - 3 = 5(x_1 - 1) + 3(x_2 - 1) + 4(x_3 - 1).$$

(iii) Let $f\colon \mathbb{R} \to \mathbb{R}^3$ be defined by

$$f(x) = \begin{pmatrix} x \\ x^2 \\ x^3 \end{pmatrix}.$$

The component functions $f^1\colon \mathbb{R} \to \mathbb{R}$, $f^2\colon \mathbb{R} \to \mathbb{R}$ and $f^3\colon \mathbb{R} \to \mathbb{R}$ are given by

$$\left.\begin{aligned} f^1(x) &= x \\ f^2(x) &= x^2 \\ f^3(x) &= x^3 \end{aligned}\right\}$$

and hence

$$\left.\begin{aligned} f_1^1(x) &= 1 \\ f_1^2(x) &= 2x \\ f_1^3(x) &= 3x^2. \end{aligned}\right\}$$

These functions are all continuous everywhere and hence f is differentiable at every point. Its derivative at -1 is the 3×1 matrix

$$f'(-1) = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}.$$

The graph of f is determined by the system of three equations

$$\left.\begin{aligned} y_1 &= x \\ y_2 &= x^2 \\ y_3 &= x^3. \end{aligned}\right\}$$

The tangent flat where $x = -1$ is given by

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix} + \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}(x+1)$$

and hence is the straight line determined by the equations

$$\left.\begin{aligned} y_1 &= x \\ y_2 &= -2x - 1 \\ y_3 &= 3x + 2. \end{aligned}\right\}$$

(iv) Let $f\colon \mathbb{R}^2 \to \mathbb{R}^2$ be defined by

$$f(x_1, x_2) = \begin{pmatrix} x_1^2 + x_2^2 \\ x_1^2 - x_2^2 \end{pmatrix}.$$

The component functions $f^1\colon \mathbb{R}^2 \to \mathbb{R}$ and $f^2\colon \mathbb{R}^2 \to \mathbb{R}$ are then given by

$$\begin{aligned} f^1(x_1, x_2) &= x_1^2 + x_2^2 \\ f^2(x_1, x_2) &= x_1^2 - x_2^2 \end{aligned}$$

and hence

$$\begin{aligned} f_1^1(x_1, x_2) &= 2x_1 \qquad f_2^1(x_1, x_2) = 2x_2 \\ f_1^2(x_1, x_2) &= 2x_1 \qquad f_2^2(x_1, x_2) = -2x_2. \end{aligned}$$

These functions are continuous everywhere and hence f is differentiable at every point. Its derivative at $(1, 1)$ is the 2×2 matrix

$$f'(1, 1) = \begin{pmatrix} 2 & 2 \\ 2 & -2 \end{pmatrix}.$$

The graph of f is determined by the two equations

$$\left.\begin{aligned} y_1 &= x_1^2 + x_2^2 \\ y_2 &= x_1^2 - x_2^2. \end{aligned}\right\}$$

The tangent flat where $(x_1, x_2) = (1, 1)$ is given by

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix} + \begin{pmatrix} 2 & 2 \\ 2 & -2 \end{pmatrix}\begin{pmatrix} x_1 - 1 \\ x_2 - 1 \end{pmatrix}$$

i.e.

$$\left.\begin{aligned} y_1 - 2 &= 2(x_1 - 1) + 2(x_2 - 1) \\ y_2 &= 2(x_1 - 1) - 2(x_2 - 1). \end{aligned}\right\}$$